



Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2012

PRELIMINARY SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 3: FINAL PRELIMINARY EXAMINATION

Mathematics Extension I

TIME ALLOWED: 2 HOURS

Outcomes Assessed	Questions	Marks
Chooses and applies appropriate arithmetic, algebraic, graphical and geometric techniques to solve problems	1, 2	34
Performs routine arithmetic and algebraic manipulation involving trigonometric identities and applies appropriate trigonometry techniques	3,4	30
Determines the derivative of a function through routine application of the rules of differentiation and relates the derivative of a function to the slope of its graph	5	12
Synthesises mathematical solutions to harder problems and communicates them in appropriate form	6	14

Question	1	2	3	4	5	6	Total	%
Marks	/15	/19	/15	/15	/12	/14	/90	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- **Each new question is to be started in a new booklet**

Question 1: BEGIN A NEW ANSWER BOOKLET

(15 marks)

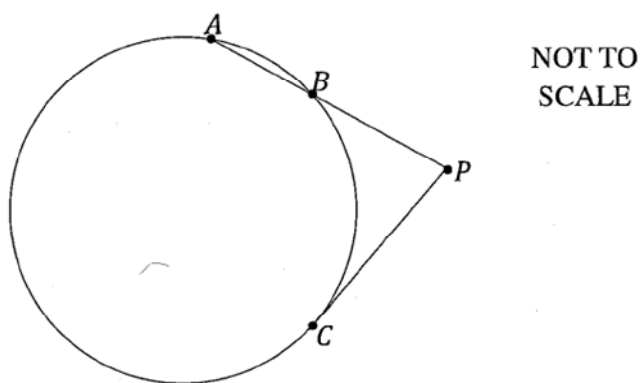
a) Simplify

i. $\frac{1}{\sqrt{(1.2)^2 + (2.1)^2}}$ (correct to 4 significant figures) 1

ii. $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ (in surd form) 1

b) Find the coordinates of the point that internally divides the interval AB in the ratio $3:2$ if A is $(-2,3)$ and B is $(3,-4)$. 2

c) In the diagram the points A, B and C lie on the circle and AB produced meets the tangent from C at the point P .



i. Given that $PC = 12, AB = 7$ and $PB = x$, find x . 2

ii. BC is the diameter of the circle passing through P, B and C . Find the length of BC . 1

d) Find the locus of the point $P(x, y)$ that moves so that P is equidistant from the points $A(3,5)$ and $B(-1,-1)$. 2

e) Solve $|x+1| = 2x-1$ for x . 3

f) Solve $\frac{2t}{1-t} \geq t$ for t . 3

Question 2: BEGIN A NEW ANSWER BOOKLET

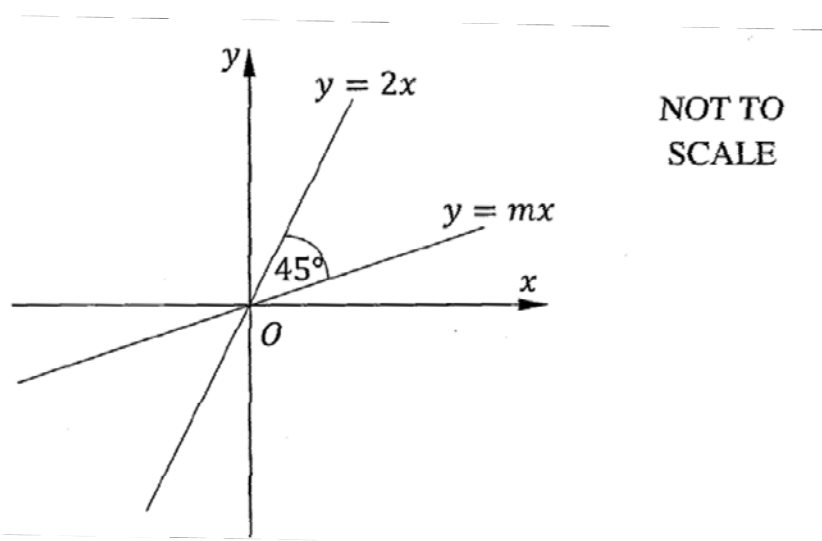
(19 marks)

a) For the curve $y = 2 + \sqrt{x-1}$:

i. Draw a neat $\frac{1}{2}$ page sketch of this curve. 2

ii. State the domain and range for this curve. 2

b) The angle between the lines $y = mx$ and $y = 2x$ is 45° , where $m > 0$, as shown in the diagram below. Find the value of m . 2



c) Find values of a, b and c such that $3x^2 - 7x - 6 \equiv a(x-2)^2 + b(x-2) + c$ for all x . 3

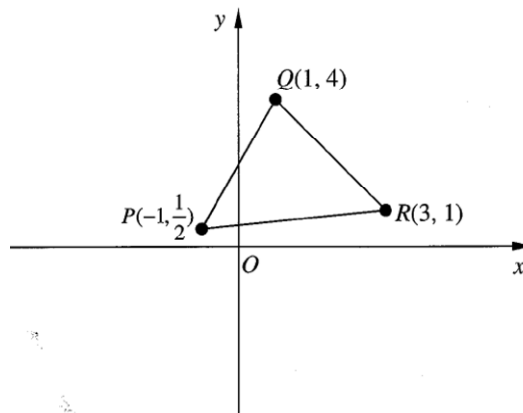
d) The polynomial $P(x) = 2x^3 - 3x^2 + x - 4$ has roots α, β and γ . Find the value of

i. $\alpha + \beta + \gamma$ 1

ii. $\alpha^2 + \beta^2 + \gamma^2$ 2

e) Find the focus and directrix of the parabola $y = 2x^2 - 4x + 7$. 3

- f) $P\left(-1, \frac{1}{2}\right)$, $Q(1, 4)$ and $R(3, 1)$ are the vertices of a triangle, as shown below.



- iii. Find the gradient of PM , where M is the midpoint of QR . 2
- iv. Show that PM is the perpendicular bisector of QR . 2

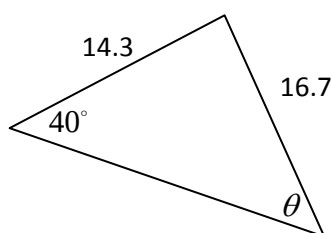
Question 3: BEGIN A NEW ANSWER BOOKLET

(15 marks)

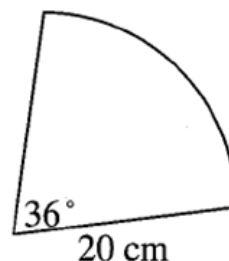
a) Give the exact value for

i. $\sin 60^\circ$ 1

ii. $\tan \frac{5\pi}{6}$ 1

b) Find the value of θ in the following triangle (to the nearest degree). 2c) Draw a neat $\frac{1}{2}$ page sketch of the curve $y = 3\sin\left(2x + \frac{\pi}{3}\right)$ for $0 \leq x \leq 2\pi$ 3

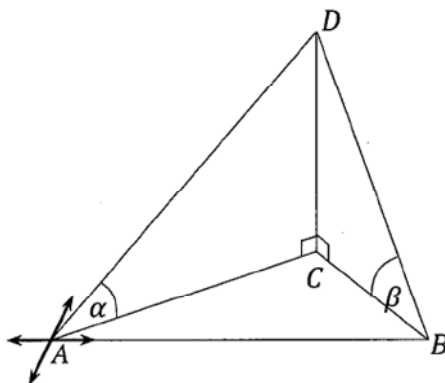
d) For the diagram given:

i. Express 36° in radian measure. 1ii. Find the exact area of the shape 2iii. Find the exact perimeter of the shape. 2e) Two cars leave the point of no return at the same time. Car A averages 80 km/h along a straight road with a bearing of $025^\circ T$ and car B averages 90 km/h along a different straight road with a bearing of $135^\circ T$.i. Draw a neat sketch showing their positions relative to the point of no return three hours after the cars leave the point of no return. 1ii. Calculate how far apart the two cars are at this time (to the nearest km). 2

Question 4: BEGIN A NEW ANSWER BOOKLET

(15 marks)

- a) A vertical pole, CD , is positioned so that the angles of elevation of the top of the pole from points A and B on the ground are α and β respectively. The ground is a level horizontal surface and the triangle ABC is right-angled at C . This information is shown in the diagram below.



If point B is due east of point A and point C is on a bearing of $060^\circ T$ from point A ,

show that $\frac{\tan \alpha}{\tan \beta} = \frac{1}{\sqrt{3}}$.

3

b) Prove $\frac{\cot \theta \cos \theta}{\cot \theta + \cos \theta} = \frac{\cos \theta}{1 + \sin \theta}$.

3

c)

i. Write down the angle expansion for $\tan(\theta - \alpha)$.

1

ii. Hence find the exact value of $\tan 165^\circ$.

2

d) Use the transformation for $r \cos(2x + \alpha)$ to express $2\sqrt{3} \cos(2x) - \sin(2x)$ as a single term. State the amplitude, period and phase shift for this expression.

3

e) Find the general solutions for $\cos 2\theta = \sin \theta$.

3

Question 5: BEGIN A NEW ANSWER BOOKLET

(12 marks)

a) Use the rules of differentiation to differentiate with respect to x :

i. $x^2 - 7x + 1$ 1

ii. $(2x^5 - 3)^4$ 2

iii. $(x^4 + 5)(x - 7)^3$ 3

b) Differentiate $f(x) = \frac{1}{x}$ from first principles. 3c) The function $f(x) = x^3 - 3x + 1$ has a root near $x = -2$. Use one iteration of Newton's Method to find a better solution. 3**Question 6: BEGIN A NEW ANSWER BOOKLET**

(14 marks)

a) Express the polynomial $P(x) = x^3 + bx^2 + cx + d$ in its factored form given it has a root at $x = 3$, a y -intercept of 3 and when $P(x)$ is divided by $(x - 2)$ it has a remainder of -3 . 4b) Using the substitution $t = \tan \frac{\theta}{2}$, solve the equation $2 \cos^2 \theta + 3 \cos \theta + 1 = 0$ for $0 \leq \theta \leq 2\pi$ 4c) Find the equation of the normal to the parabola $y = x^2 + 4x - 5$ at the point $(-1, -8)$ and hence find the co-ordinates of the point where it again meets the parabola. 3d) Solve $\cos\left(5x - \frac{\pi}{3}\right) = \sin 3x$ for $0 \leq x \leq 2\pi$. 3

Question 1:

a)

i) 0.4134

ii) $\frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}$

$$= \frac{(\sqrt{2}-1)^2}{2-1}$$

$$= 2 - 2\sqrt{2} + 1$$

$$= 3 - 2\sqrt{2}$$

b) $x = \frac{mx_2 + nx_1}{m+n} \quad y = \frac{my_2 + ny_1}{m+n}$
$$= \frac{3 \times 3 + 2 \times -2}{3+2} \quad = \frac{3 \times -4 + 2 \times 3}{3+2}$$

$$= \frac{5}{5}$$

$$= 1$$

$$= \frac{-6}{5}$$

$$= -1\frac{1}{5}$$

Hence point is $\left(1, -1\frac{1}{5}\right)$

c)

i) $AP \cdot PB = CP^2$

$$x(x+7) = 12^2$$

$$x^2 + 7x = 144$$

$$x^2 + 7x - 144 = 0$$

$$(x+16)(x-9) = 0$$

$$x = -16, 9$$

Reject -16 as x is a length, so x=9.

ii) BC a diameter

$$\Rightarrow \angle BPC = 90^\circ (\angle \text{ in semicircle })$$

Hence:

$$BC^2 = x^2 + 12^2$$

$$= 225$$

$$BC = 15$$

d) For $|PA| = |PB|$ where

$$|PA|^2 = (x-3)^2 + (y-5)^2 \text{ and}$$

$$|PB|^2 = (x+1)^2 + (y+1)^2$$

MARKING

COMMENTS

① answer with
sig figs correct• Many wrote 0.413 -
the 0 before the dp is
not significant.

① answer

• Many failed to
simplify correctly① correct subst in
formula

① correct point

① correct circle
property and
equation• This formula was
poorly known.

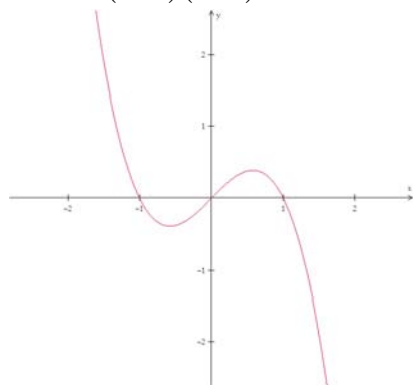
① answer

① answer

① correct set-up

	MARKING	COMMENTS
$(x-3)^2 + (y-5)^2 = (x+1)^2 + (y+1)^2$ $x^2 - 6x + 9 + y^2 - 10y + 25 = x^2 + 2x + 1 + y^2 + 2y + 1$ $8x + 12y - 32 = 0$ $2x + 3y - 8 = 0$		
e) Two cases: $x+1 = 2x-1 \quad -(x+1) = 2x-1 =$ $2 = x \quad 0 = 3x$ <i>check :</i> $LHS = 2+1 $ <i>check :</i> $LHS = 0+1 $ $= 3 \quad = 1$ $RHS = 4-1 \quad RHS = 0-1$ $= 3 \quad = -1$ $= LHS \quad \neq LHS$ $(accept) \quad (reject)$	1 correct equation 1 cases correct 1 checking done	- Full marks given for $8x + 12y - 32 = 0$ but students should realise that any answer is expected to be in simplest form. - Many failed to test cases.
Hence $x = 2$ is the only solution. f) $\frac{2t}{1-t} \geq t$ $(1-t)^2 \frac{2t}{(1-t)} \geq t(1-t)^2$ $2t(1-t) \geq t(1-t)^2$ $2t(1-t) - t(1-t)^2 \geq 0$ $(1-t)[2t - t(1-t)] \geq 0$ $(1-t)(2t - t + t^2) \geq 0$ $(1-t)(t + t^2) \geq 0$ $t(1-t)(1+t) \geq 0$	1 answer 1 method set-up correctly 1 justification and/or evidence to support answer 1 answer	- Many algebraic errors even when students know the correct method. - Mark deducted of equality sign for $t < 1$.

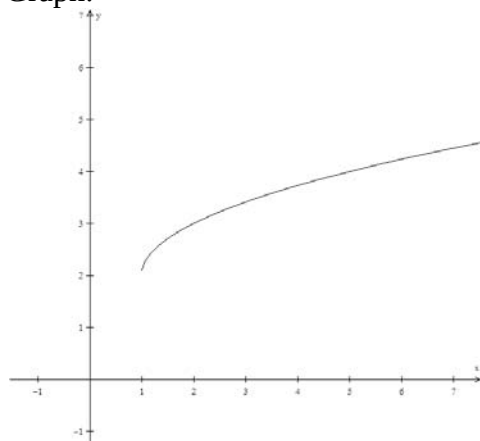
Noting $t \neq 1$ from original expression, hence $t \leq -1$ and $0 \leq t < 1$.



Question 2:

a) For $y = 2 + \sqrt{x-1}$

i) Graph:

ii) Domain: $x \geq 1$ Range: $y \geq 2$

b) $\tan \theta = \frac{|m_2 - m_1|}{1 + m_1 m_2}$

$$\tan 45 = \frac{|m - 2|}{1 + 2m}$$

$$1 + 2m = |m - 2|$$

Two cases:

$$1 + 2m = m - 2 \quad 1 + 2m = -(m - 2)$$

$$m = -3$$

$$3m = 1$$

reject as $m > 0$

$$m = \frac{1}{3}$$

$$\therefore m = \frac{1}{3}$$

$$\begin{aligned} \text{c) } RHS &= a(x^2 - 4x + 4) + bx - 2b + c \\ &= ax^2 - 4ax + 4a + bx - 2b + c \\ &= ax^2 + (b - 4a)x + 4a - 2b + c \end{aligned}$$

Equating co-efficients:

$$x^2: \quad a = 3$$

$$x: \quad -7 = b - 4a$$

$$b = 5$$

$$\text{const:} \quad -6 = 4a - 2b + c$$

$$c = -8$$

$$\text{i.e. } a = 3, b = 5, c = -8$$

MARKING

COMMENTS

① basic shape & direction correct

① vertex correct

① domain correct

① range correct

① correct use of angle between two lines formula

① correct answer with justification

① correct expansion leading to correct value for a ① b correct① c correct

- Many could not identify and sketch the basic $\sqrt{\quad}$ equation & graph

- This critical point for the curve was often poorly done.

- Well done for those who graphed correctly.

- Many did not use the 2 cases from the absolute value.

- Some did not indicate the condition $m > 0$ and thus get the correct m value.

- Many perfect square expansion errors!

- Many sign errors when calculation the b value

- The constant term often was missing one term.

	MARKING	COMMENTS
d) For $P(x) = 2x^3 - 3x^2 + x - 4$ i) $\alpha + \beta + \gamma = \frac{-b}{a}$ $= \frac{3}{2}$ ii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$ $= \left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right)$ $= \left(\frac{3}{2}\right)^2 - 2\left(\frac{1}{2}\right)$ $= 1\frac{1}{4}$ e) $y = 2x^2 - 4x + 7$ $y = 2(x^2 - 2x + 1 - 1) + 7$ $= 2(x-1)^2 + 5$ $\frac{1}{2}(y-5) = (x-1)^2$ So Vertex is at $(1, 5)$ and $4a = \frac{1}{2}$ $a = \frac{1}{8}$ Hence focus is $\left(1, 5\frac{1}{8}\right)$ and directrix is $y = 4\frac{7}{8}$ f) i) Midpoint is $\left(\frac{1+3}{2}, \frac{4+1}{2}\right)$ or $\left(2, \frac{5}{2}\right)$ Hence gradient of PM: $m = \frac{\frac{5}{2} - 1}{2 - 1}$ $= \frac{2}{3}$ ii) Gradient QR: $m = \frac{4-1}{1-3}$ $= \frac{3}{-2}$ Hence $m_{PM} \times m_{QR} = \frac{2}{3} \times \frac{3}{-2}$ $= -1$	1 answer 1 correct three term expansion for a square 1 answer 1 correct completion of the squares 1 focus correct 1 directrix correct 1 midpoint correct 1 gradient correct 1 shows perpendicular	• Generally well done. • Mostly well done, but some had $-2\alpha\beta\gamma$ instead of the paired roots. • Many simple calculation errors. • The factor of 2 caused many issues with incorrect algebra. • Many used the a value to adjust the x value rather than the y value! • This is a point! • This is an equation (line)! • Generally well done. • Generally well done. • Generally well done.

So as M is the midpoint of QR (from i) and $PM \perp QR$, then PM is the perpendicular bisector of QR (as reqd).

Question 3:

a) Exact values are

i) $\sin 60^\circ$

$$= \frac{\sqrt{3}}{2}$$

ii) $\tan \frac{5\pi}{6}$

$$= -\tan \frac{\pi}{6}$$

$$= -\frac{1}{\sqrt{3}}$$

b) $\frac{\sin \theta}{14.3} = \frac{\sin 40^\circ}{16.7}$

$$\sin \theta = \frac{14.3 \sin 40^\circ}{16.7}$$

$$\theta = \sin^{-1} \left(\frac{14.3 \sin 40^\circ}{16.7} \right)$$

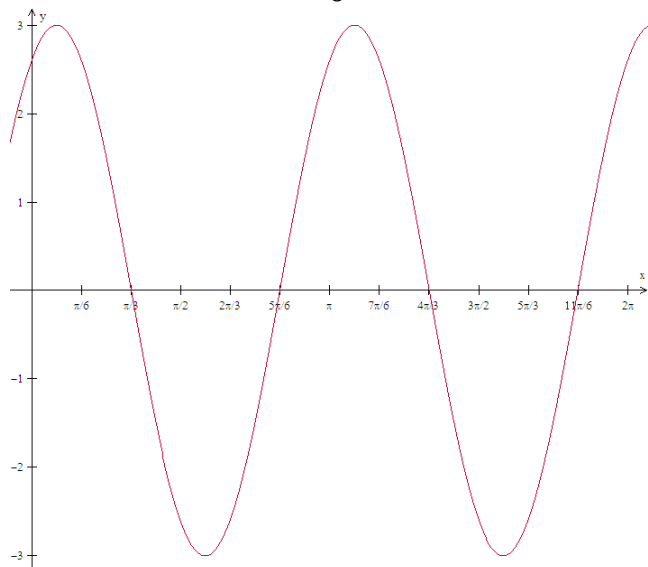
$$= 33.39521023$$

$$\approx 33^\circ \text{ (nearest } ^\circ \text{)}$$

c) $y = 3 \sin \left(2x + \frac{\pi}{3} \right)$: Amplitude is 2, period is

$$\pi \text{ and shifted } 0 = 2x + \frac{\pi}{3} \text{ (i.e. to the left).}$$

$$x = -\frac{\pi}{6}$$

**MARKING**

① conclusion linked correctly

① answer

① answer

① sine rule correct with substitutions

① answer

① amplitude correct

① period correct

① phase shift correct

COMMENTS

- Many either missed or went to great lengths showing the bisection component.

- Some students used composite angle expansion rather than basic trig identities.

- Generally well done.

- Poorly attempted – many did not factor the 2 out to find the correct shift.

- Some stated the correct period, but then did not reflect this correctly in their graph.

- Many need to take greater care graphing.

d)

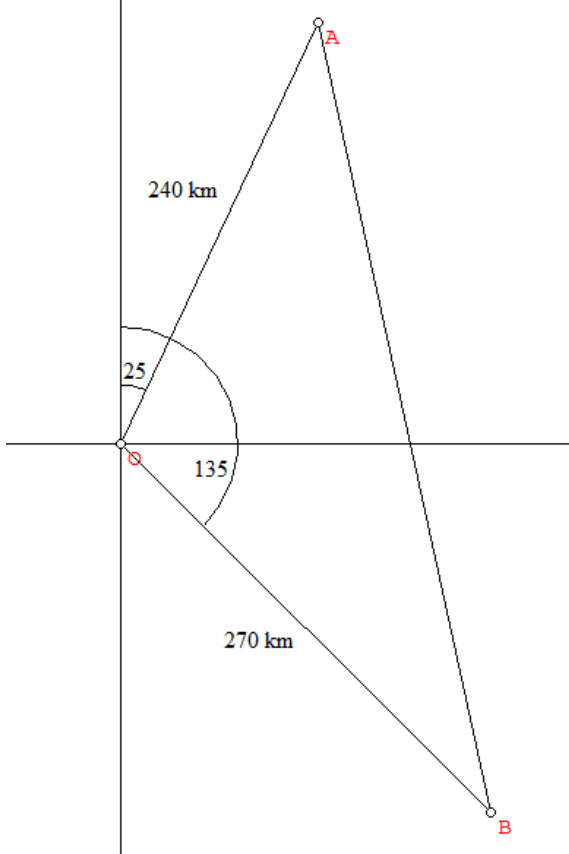
$$\begin{aligned} \text{i) } 36 \times \frac{\pi}{180} \\ = \frac{\pi}{5} \end{aligned}$$

$$\begin{aligned} \text{ii) } A &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} \times 20^2 \times \frac{\pi}{5} \\ &= 40\pi u^2 \end{aligned}$$

$$\begin{aligned} \text{iii) } P &= 2r + r\theta \\ &= 2 \times 20 + 20 \times \frac{\pi}{5} \\ &= 40 + 4\pi \end{aligned}$$

e)

i) Diagram:



$$\begin{aligned} \text{ii) } \angle AOB &= 135 - 25 \\ &= 110 \end{aligned}$$

Hence

$$\begin{aligned} d^2 &= 240^2 + 270^2 - 2 \times 240 \times 270 \times \cos 110^\circ \\ &= 174825.8106 \\ d &= 418.1217652 \\ &\approx 418 \text{ km} \end{aligned}$$

MARKING

COMMENTS

① answer

① correct formula
& substitution

① answer

① answer
① exact value
correct① diagram
correct① cosine rule &
substitutions
correct

① answer

- Many students did not know the radian formulae or could not simplify their answer.

- Many careless errors including calculator in radians, and incorrect distances OA, OB

Question 4:

- a) In $\triangle ABC$ (linking $\triangle$): with $\angle C = 90^\circ$ and $\angle CAB = 90 - 60$ (from bearing), then

$$\tan 30^\circ = \frac{BC}{AC}$$

$$\therefore \frac{BC}{AC} = \frac{1}{\sqrt{3}}$$

In $\triangle ADC$:

$$\tan \alpha = \frac{CD}{AC}$$

In $\triangle ADB$:

$$\tan \beta = \frac{CD}{BC}$$

Hence:

$$\begin{aligned} \frac{\tan \alpha}{\tan \beta} &= \frac{\frac{CD}{AC}}{\frac{CD}{BC}} \\ &= \frac{BC}{AC} \\ &= \frac{1}{\sqrt{3}} \text{ as reqd.} \end{aligned}$$

b)

$$\begin{aligned} LHS &= \frac{\frac{\cos \theta}{\sin \theta} \cdot \cos \theta}{\frac{\cos \theta}{\sin \theta} + \cos \theta} \\ &= \frac{\frac{\cos^2 \theta}{\sin \theta}}{\frac{\cos \theta + \cos \theta \sin \theta}{\sin \theta}} \\ &= \frac{\cos^2 \theta}{\sin \theta} \cdot \frac{\sin \theta}{\cos \theta (1 + \sin \theta)} \\ &= \frac{\cos \theta}{(1 + \sin \theta)} \\ &= RHS \end{aligned}$$

c)

i) $\tan(\theta - \alpha) = \frac{\tan \theta - \tan \alpha}{1 + \tan \theta \tan \alpha}$

MARKING

❶ set-up for linking triangle

❶ tan relationships in side triangles

❶ algebra to reqd solution correct.

❶ sin/cos substitutions correct

❶ fraction manipulations correct

❶ LHS...RHS format correct

❶ answer

COMMENTS

Students who allocated exact lengths in $\triangle ABC$ were awarded 1 mark.

Mostly well done.

ii)	MARKING	COMMENTS
$\begin{aligned}\tan 165^\circ &= -\tan 15^\circ \\ &= -\tan(60 - 45) \\ &= -\frac{\tan 60 - \tan 45}{1 + \tan 60 \tan 45} \\ &= -\frac{\sqrt{3} - 1}{1 + \sqrt{3}} \left(\times \frac{\sqrt{3} - 1}{\sqrt{3} - 1} \right) \\ &= -\frac{(\sqrt{3} - 1)^2}{3 - 1} \\ &= -\frac{1}{2}(3 - 2\sqrt{3} + 1) \\ &= \sqrt{3} - 2\end{aligned}$	① use of previous part (can use 45 and 30, but resulting fractions are more complex) ① answer	Essential to use tan of difference (the “Hence” in this part); marks were not awarded for using the sum You must rationalize the denominator
d) $r \cos(2x + \alpha) = r \cos 2x \cos \alpha - r \sin 2x \sin \alpha$ With $r \cos \alpha = 2\sqrt{3}$ and $r \sin \alpha = 1$, then $\frac{r \sin \alpha}{r \cos \alpha} = \frac{1}{2\sqrt{3}}$ $\tan \alpha = \frac{1}{2\sqrt{3}}$ $\alpha = \tan^{-1} \frac{1}{2\sqrt{3}}$ $\approx 16.1^\circ$ $r^2 (\sin^2 \alpha + \cos^2 \alpha) = 1^2 + (2\sqrt{3})^2$ $r^2 = 13$ $r = \sqrt{13}$ Hence amplitude $= \sqrt{13}$, period is $\frac{2\pi}{2}$ or π and the phase shift is $\frac{1}{2} \tan^{-1} \left(\frac{1}{2\sqrt{3}} \right)$ which is $\approx 8.05^\circ$ (or 0.14°).	① finds α ① finds r ① interprets & gives values correctly	Phase shift with $2x$ not interpreted well
e) $\cos 2\theta = \sin \theta$ $1 - 2\sin^2 \theta = \sin \theta$ $0 = 2\sin^2 \theta + \sin \theta - 1$ $= 2\sin^2 \theta + 2\sin \theta - \sin \theta - 1$ $= 2\sin \theta (\sin \theta + 1) - (\sin \theta + 1)$ $= (\sin \theta + 1)(2\sin \theta - 1)$	① getting quadratic equation	

Hence:

$$(\sin \theta + 1) = 0$$

$$\sin \theta = -1$$

$$\theta = n\pi + (-1)^n \sin^{-1}(-1)$$

Two cases: odd and even:

Even case: $n = 2m$

$$\theta = 2m\pi + \frac{\pi}{2}$$

$$= (4m+1)\frac{\pi}{2}$$

Odd case: $n = 2m+1$

$$\theta = (2m+1)\pi - \frac{\pi}{2}$$

$$= (4m+2-1)\frac{\pi}{2}$$

$$= (4m+1)\frac{\pi}{2}$$

$$(2\sin \theta - 1) = 0$$

$$2\sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{2}\right)$$

$$= n\pi + (-1)^n \left(\frac{\pi}{6}\right)$$

Two cases: odd and even:

Even case:

$$n = 2m:$$

$$\theta = 2m\pi + \frac{\pi}{6}$$

$$= (12m+1)\frac{\pi}{6}$$

Odd case:

$$n = 2m+1:$$

$$\theta = (2m+1)\pi - \frac{\pi}{6}$$

$$= (12m+5)\frac{\pi}{6}$$

MARKING

COMMENTS

1 answer

1 generator
correctMarks were awarded for
appropriate general
solutions that generated
correct angles.Alternative
Method Marks:

1 set-up correct

Alternative (simpler) method:

$$\cos 2\theta = \cos\left(\frac{\pi}{2} - \theta\right)$$

$$\cos 2\theta = c = \cos\left(\frac{\pi}{2} - \theta\right)$$

Gives 2 equations:

$$2\theta = 2n\pi \pm \cos^{-1} c \quad \frac{\pi}{2} - \theta = \cos^{-1} c$$

Combining:

$$2\theta = 2n\pi \pm \left(\frac{\pi}{2} - \theta\right)$$

Negative case:

$$2\theta = 2n\pi - \left(\frac{\pi}{2} - \theta\right)$$

$$\theta = (4n-1)\frac{\pi}{2}$$

$$\left(\frac{3\pi}{2}, \frac{7\pi}{2}, \frac{11\pi}{2}, \dots\right)$$

Positive case:

$$2\theta = 2n\pi + \frac{\pi}{2} - \theta$$

$$3\theta = (4n+1)\frac{\pi}{2}$$

$$\theta = (4n+1)\frac{\pi}{6}$$

Note: this positive case generates the solutions:

$$\frac{\pi}{6}, \frac{5\pi}{6}, \frac{9\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{21\pi}{6} \dots \text{ which include}$$

the negative case solutions

$$\left(\frac{9\pi}{6} \equiv \frac{3\pi}{2}; \frac{21\pi}{6} \equiv \frac{7\pi}{2}, \text{etc...}\right) \text{ thus the general}$$

solution is $\theta = (4n+1)\frac{\pi}{6}, n = 0, \pm 1, \pm 2, \dots$

MARKING

COMMENTS

❶ negative case
generator correct

❶ positive case
generator correct

Also Note:

The sin generators from the first method give:

$$(12m+1)\frac{\pi}{6} \text{ gives } \frac{\pi}{6}, \frac{13\pi}{6}, \frac{25\pi}{6}, \dots,$$

$$(12m+5)\frac{\pi}{6} \text{ gives } \frac{5\pi}{6}, \frac{17\pi}{6}, \frac{29\pi}{6}, \dots \text{ and with}$$

the $\theta = (4n+1)\frac{\pi}{2}$ generator common to both methods, these three generate every third term

of the solution $\theta = (4n+1)\frac{\pi}{6}, n = 0, \pm 1, \pm 2, \dots$

MARKING

COMMENTS

Question 5:

a)

i) $2x - 7$

ii) $4(2x^5 - 3)^3 \cdot 10x^4$
 $= 40x^4(2x^5 - 3)^3$

iii) $u = x^4 + 5 \quad v = (x - 7)^3$
 $du = 4x^3 \quad dv = 3(x - 7)^2$

$$\frac{dy}{dx} = u \cdot dv + v \cdot du$$
$$= (x^4 + 5) \cdot 3(x - 7)^2 + (x - 7)^3 \cdot 4x^3$$
$$= (x - 7)^2 [3(x^4 + 5) + 4x^3(x - 7)]$$
$$= (x - 7)^2 (7x^4 - 28x^3 + 15)$$

b) $\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$

$$= \lim_{\delta x \rightarrow 0} \frac{\frac{1}{x + \delta x} - \frac{1}{x}}{\delta x}$$

$$= \lim_{\delta x \rightarrow 0} \frac{\frac{x - (x + \delta x)}{x(x + \delta x)}}{\delta x}$$

$$= \lim_{\delta x \rightarrow 0} \frac{x - x - \delta x}{x\delta x(x + \delta x)}$$

$$= \lim_{\delta x \rightarrow 0} \frac{-\delta x}{x\delta x(x + \delta x)}$$

$$= \lim_{\delta x \rightarrow 0} \frac{-1}{x(x + \delta x)}$$

$$= \frac{-1}{x^2}$$

c) $f(-2) = (-2)^3 - 3(-2) + 1$
 $= -1$

$f'(x) = 3x^2 - 3$

$f'(-2) = 3(-2)^2 - 3$
 $= 9$

MARKING

COMMENTS

① answer

① chain rule
correct

① answer

① set-up correct

① product rule
correctly
substituted① answer in
simplest form① first principles
formula with
correct
substitution① algebra
correct to be able
to cancel δx ① limit resolved
correctly① values for
 $x = -2$ correct• This question
generally well done.

Hence

$$\begin{aligned}
 x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\
 &= -2 - \frac{-1}{9} \\
 &= -1\frac{8}{9}
 \end{aligned}$$

Question 6:a) For $P(x) = x^3 + bx^2 + cx + d$ y-intercept of 3 $\Rightarrow d = 3$ root at $x = 3 \Rightarrow P(3) = 0$

$$0 = 27 + 9b + 3c + d$$

$$-30 = 9b + 3c \quad \text{①}$$

Remainder of -3 when division by $(x - 2)$

$$\Rightarrow P(2) = -3$$

$$-3 = 8 + 4b + 2c + d$$

$$-14 = 4b + 2c \quad \text{②}$$

Eliminating c :

$$\text{③} = \text{①} \times 2 : 18b + 6c = -60$$

$$\text{④} = \text{②} \times 3 : 12b + 6c = -42$$

$$\text{③} - \text{④} : 6b = -18$$

$$b = -3$$

In ①: $c = -1$

$$\therefore P(x) = x^3 - 3x^2 - x + 3$$

$$\begin{array}{r}
 x^2 - 1 \\
 (x - 3) \overline{) x^3 - 3x^2 - x + 3} \\
 \underline{x^3 - 3x^2} \\
 -x + 3 \\
 \underline{-x + 3} \\
 0
 \end{array}$$

Hence

$$P(x) = (x - 3)(x - 1)(x + 1)$$

MARKING

COMMENTS

① formula
correct

① answer

① interprets
given data
correctly to give
initial equations① forms
equations to
solve for b, c
correctly① values for b, c
correct① poly division
or equivalent to
get answer in
factored form

- The main error was getting the function and derivative values in the wrong order.

Many failed to give $P(x)$ in factored form as required.

b) Substituting: $0 = 2\left(\frac{1-t^2}{1+t^2}\right)^2 + 3\left(\frac{1-t^2}{1+t^2}\right) + 1$

$$\begin{aligned} 0 &= 2(1-t^2)^2 + 3(1-t^2)(1+t^2) + (1+t^2)^2 \\ &= 2(1-2t^2+t^4) + 3(1-t^4) + 1+2t^2+t^4 \\ &= 2-4t^2+2t^4+3-3t^4+1+2t^2+t^4 \\ &= 6-2t^2 \end{aligned}$$

$$3 = t^2$$

$$t = \pm\sqrt{3}$$

$$\therefore \tan\left(\frac{\theta}{2}\right) = \pm\sqrt{3}$$

$$\frac{\theta}{2} = \tan^{-1}(\pm\sqrt{3})$$

$$= \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

$$\theta = \frac{2\pi}{3}, \frac{4\pi}{3} \quad (0 \leq \theta \leq 2\pi)$$

Checking $\theta = \pi$ in original equation:

$$LHS = 2\cos^2 \pi + 3\cos \pi + 1$$

$$= 2(-1)^2 + 3(-1) + 1$$

$$= 0$$

$$= RHS$$

$\therefore \theta = \pi$ is also a solution

$$\therefore \theta = \frac{2\pi}{3}, \pi, \frac{4\pi}{3} \quad (0 \leq \theta \leq 2\pi)$$

c) $y' = 2x + 4$

@ $x = -1$: $y' = 2$

So normal has gradient $-\frac{1}{2}$

$$y - 8 = -\frac{1}{2}(x - 1)$$

$$2y + 16 = -x - 1$$

$$x + 2y + 17 = 0$$

is the equation of the normal.

Then, re-arranging:

$$2y = -x - 17$$

And the parabola becomes ($\times 2$):

$$2y = 2x^2 + 8x - 10, \text{ so adding:}$$

$$0 = 2x^2 + 9x + 7$$

$$= 2x^2 + 2x + 7x + 7$$

$$= 2x(x+1) + 7(x+1)$$

MARKING

❶ correct t substitutions

❶ correct t values

❶ correct θ values

❶ Checks and includes $\theta = \pi$ as a solution

❶ equation of normal

COMMENTS

Many students need to learn the ' t ' results; these had to be used as specified in the question.

The $\pm$ was frequently missed.

Some students did not check $\theta = \pi$.

Too many careless errors in this part.

	MARKING	COMMENTS
$0 = (x+1)(2x+7)$ So $x = -1, \frac{-7}{2}$ As $(-1, -8)$ is the original point, the other point has a y value $y = \left(\frac{-7}{2}\right)^2 + 4\left(\frac{-7}{2}\right) - 5$ $= \frac{-27}{4}$ or $\left(-3\frac{1}{2}, -6\frac{3}{4}\right)$ d) $\cos\left(5x - \frac{\pi}{3}\right) = \sin(3x)$ $\cos\left(5x - \frac{\pi}{3}\right) = c = \cos\left(\frac{\pi}{2} - 3x\right)$ $\therefore 5x - \frac{\pi}{3} = 2n\pi \pm \left(\frac{\pi}{2} - 3x\right)$ Positive case: $5x - \frac{\pi}{3} = 2n\pi + \frac{\pi}{2} - 3x$ $8x = 2n\pi + \frac{5\pi}{6}$ $= (12n+5)\frac{\pi}{6}$ $x = (12n+5)\frac{\pi}{48}$ $= \frac{5\pi}{48}, \frac{17\pi}{48}, \frac{29\pi}{48}, \frac{41\pi}{48},$ $\frac{53\pi}{48}, \frac{65\pi}{48}, \frac{77\pi}{48}, \frac{89\pi}{48}$ Negative case: $5x - \frac{\pi}{3} = 2n\pi - \frac{\pi}{2} + 3x$ $2x = 2n\pi - \frac{\pi}{6}$ $= (12n-1)\frac{\pi}{6}$ $x = (12n-1)\frac{\pi}{12}$ $= \frac{11\pi}{12}, \frac{23\pi}{12}$	1 solves equations for x values 1 point correct 1 resolves to cases (or equivalent) 1 solutions from one case (or equivalent) correct 1 solutions from other case (or equivalent) correct	Many failed to realize that $x=1$ had to be one of the solutions – given in the question! Poorly done – too many students did not use general solutions to generate all the solutions in the given domain.